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The effect of early physiotherapy on the recovery of mandibular function after orthognathic surgery for class III correction. Part II: Electromyographic activity of masticatory muscles

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ABSTRACT

The study was conducted to evaluate the effect of early physical rehabilitation by comparing the differences of surface electromyographic (sEMG) activity in the masseter and anterior temporalis muscles after surgical correction of skeletal class III malocclusion. The prospective study included 63 patients; the experimental groups contained 31 patients who received early systematic physical rehabilitation; the control group (32 patients) did not receive physiotherapy. The amplitude of sEMG in the masticatory muscles reached 72.6–121.3% and 37.5–64.6% of pre-surgical values in the experimental and control groups respectively at 6 weeks after orthognathic surgery (OGS). At 6 months after OGS, the sEMG reached 135.1–233.4% and 89.6–122.5% of pre-surgical values in the experimental and control groups respectively. Most variables in the sEMG examination indicated that recovery of the masticatory muscles in the experimental group was better than the control group as estimated in the early phase (T1 to T2) and the total phase (T1 to T3); there were no significant differences between the mean recovery percentages in the later phase (T2 to T3). Early physical rehabilitative therapy is helpful for early recovery of muscle activity in masticatory muscles after OGS. After termination of physical therapy, no significant difference in recovery was indicated in patients with or without early physiotherapy.

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1. Introduction

Orthognathic surgery (OGS) can provide predictable changes to the jaw bones and intermaxillary occlusal relations for correcting dentofacial deformities. Outcome assessments of the treatment are not only based on aesthetic improvement and surgical stability, but also on the oral functional changes that can be assessed by patient-centered reporting of long-term quality of life (Pahkala and Kellakoski, 2007).

Patients with skeletal class III malocclusion usually have limitation in masticatory function. The masticatory system is considered as a force/moment system which obeys the laws of mechanics. In theory, by correcting craniofacial deformities, it is possible to influence the maximum bite force of the patients (Finn et al., 1980; Throckmorton et al., 1980). In patients with mandibular prognathism, a longer bite force moment arm is likely to decrease their maximum bite force. Surgical setback of the mandible has mechanical advantages and surgical correction of mandibular prognathism is expected to increase muscle efficiency and maximum bite force (Throckmorton et al., 1995). In patients with mandibular prognathism, EMG during maximal isometric clenching was reported to have increased one year after surgery.

However, there are many clinical and biological complications which can affect muscles following OGS including: muscular

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atrophy and denervation; decreased muscular strength and extensibility; and decreased muscle mass and length (Storum and Bell, 1984; Aragon and Van Sickels, 1987). Researchers have suggested the use of programmed rehabilitation after OGS (Bell et al., 1983). The physiotherapy could improve muscle vascularity, increase muscle mass, decrease muscle fatigability and increase muscle strength (Boyd and Karas, 1991). Thus patients with class III correction may need effective physiotherapy to reduce these complications and aid recovery of muscle function.

Surface electromyography (sEMG) is a non-invasive measurement technique, which reflects the degree of myoelectrical activity (Basmajian and De Luca, 1985). Due to their accessibility, the anterior temporalis and the superficial masseter muscles are the most common sites used for sEMG recording of the muscles of mastication. EMG is frequently recorded during clenching at maximal or submaximal bite force and during chewing. The quantity of sEMG reflects the level of muscle activity. Generally, this EMG/bite-force relationship appears to be linear (Bakke and Paulsen, 1989; Throckmorton, 2006). Under isometric constriction, the sEMG value of the anterior temporalis and superficial masseter muscles increases with bite force. Therefore sEMG provides a practical objective tool for the assessment of masticatory function.

Our previous study investigated the changes over time of masticatory electromyographic activity in OGS patients with skeletal class III malocclusion. The serial sEMG data indicated a significant decrease in activity from before surgery to one month after OGS, and then recovered from one month onwards. No significant difference of EMG activity between the initial and final measurements was noted (Ko et al., 2013). However, factors which could accelerate or decelerate the functional recovery are not well known. The role of physical rehabilitation on the recovery of masticatory muscle activity after OGS needs further clinical investigation.

The aim of this study is to evaluate the effect of early rehabilitative physiotherapy by comparing the longitudinal changes in myoelectrical activity of the anterior temporalis and superficial masseter muscles at three different time intervals in patients with orthognathic surgical correction of skeletal class III malocclusion.

2. Material and methods

2.1. Subject sample

The 63 subjects in this prospective study consisted of two groups of patients. The experimental group included 31 patients with skeletal class III malocclusion who received two-jaw surgery as well as early physical rehabilitation and dietary instruction; 9 male and 22 female, with a mean age of 24.0 years (SD 3.6). The control group consisted of 32 patients who received two-jaw surgery without any physical rehabilitation and dietary instruction; 8 male and 24 female, with a mean age of 25.3 years (SD 4.8). Two orthodontists, one for the experimental group and the other for the control group treated these consecutive series of patients.

The inclusion criteria for the subjects included: 1) skeletal class III dentofacial deformities with ANB angle $<0^{\circ}$; 2) they required two-jaw OGS (BSSO and Le Fort 1) and orthodontic treatment; 3) absence of temporomandibular joint pain and preoperative opening mouth limitation; 4) absence of craniofacial anomalies and abnormal cranial form (extreme brachycephalic, dolichocephalic or plagiocephalic); 5) absence of facial trauma, mandibular condylar fracture or jaw bone defects; 6) presence of dentition: all six front teeth in both sides of the jaw, with at least 1 M in each quadrant; 7) complete diagnostic data and records.

For OGS, mandibular osteotomy with a modification of the Hunsuck intraoral sagittal split osteotomy of the ramus (Hunsuck, 1968) was used. Detachment of the internal pterygoid muscles was performed during osteotomy. Three 2-hole miniplates and six monocortical screws were used on each side of mandible for fixation. In all patients, no intermaxillary fixation was applied after surgery. Two main surgeons with similar surgical techniques performed the OGS. Postsurgical orthodontic treatment was completed for detailing the alignment and interdigitation in both groups.

The patient records included demographic data, lateral and posteroanterior cephalometric radiographs before treatment, and serial examination of sEMG in the anterior temporalis and masseter muscles. The time points of sEMG records were obtained: before OGS (T1); six weeks after OGS (T2); and six months after OGS (T3). The average timing of T2 was 45.3 ± 4.5 days after OGS in the experimental group, and 48.1 ± 6.1 in the control group. The timing for T3 was 209.7 ± 11.9 days after OGS in the experimental group and 208.5 ± 14.3 in the control group. No significant difference in the timing of the two groups was found.

2.2. Physical rehabilitation and dietary instruction

The physical rehabilitation started on the 8th day after OGS. Patients were required to practice this physical rehabilitation once before every meal, at least three times each day. Following each session of active jaw exercise, patients could relax and stretch masticatory muscles by manual massage. In the 5th week, patients were instructed to use passive jaw exercise with the aid of a tongue blade; after that, a set of three isometric clenches of the masticatory muscles, holding 10 s each time were performed. The proposals of early physiotherapy and dietary instruction accepted by the experimental group are shown in Table 1 (Teng et al., 2015).

To remind subjects to practice their physical rehabilitation exercises and follow dietary instruction, every subject in the experimental group received a rehabilitation diary after pre-surgical instruction on the physiotherapy and diet. The rehabilitation diary contained a check list and notes of the post-operative physical rehabilitation plan and dietary recommendations.

2.3. The surface electromyography of masticatory muscles

The sEMG activity was tested on the bilateral anterior temporalis and masseter muscles. The recordings were obtained at three time points: T1, before surgery; T2, 6 weeks after surgery; and T3, 6

Table 1
The proposals of early physiotherapy and dietary instruction for the experimental group.

Time after orthognathic surgery	Physiotherapy	Dietary instruction
Week 1 (0–7 days)	Avoid jaw movement.	Liquid diet
Week 2 (8–14 days) Week 3 (15–21 days) Week 4 (22–28 days)	6 times active maximal mouth opening, holding 30 s each time; 10 times lateral excursion movement, holding 5 s each time; 10 times protrusive and retrusive movement, holding 5 s each time.	Soft diet
Week 5 (29–35 days) Week 6 (36–42 days)	6 times passive maximal mouth opening assisted by use of a tongue blade, 10 times active lateral excursion, 10 times protrusive and retrusive jaw movement, and 3 times isometric biting exercise of masticatory muscles, holding 10 s per time.	Normal diet

months after surgery. The registration of the surface electrodes and manipulation of sEMG (Zebris EMG 4, Zebris GmbH, Isny im Allgäu, Germany) are described previously (Ko et al., 2013). The location of the EMG electrodes is shown in Fig. 1.

To determine the maximum voluntary contraction (MVC), a bite test was conducted first in habitual intercuspation and then on cotton pads. Patients were asked to bite as hard as possible without relaxing for 2 s. The patients relaxed again and opened their mouths slightly, after which the recording was stopped. The same procedure was repeated with a bilateral cotton roll (10 mm in diameter) bite on bilateral posterior teeth. To determine the firing pattern, the patients were asked to press their teeth together with increasing force for 10 s. The patients relaxed again and opened their mouths slightly, after which the recording was stopped.

The EMG signals were processed by the software program (WinJaw® 10.5, Zebris GmbH, Isny im Allgäu, Germany) in the time domain by the root mean square calculation and expressed in microvolt units (μV). The definitions and measurement of parameters of myoelectrical activity of the masseter and anterior temporalis muscles are listed in Table 2.

2.4. Statistical analysis

The differences in age and sex distribution between the two groups were compared using Student's *t*-test and Pearson chi-square test, respectively. The percentage of recovery in myoelectrical activity of the masseter and anterior temporalis muscles was defined as the value of each parameter at T2 and T3 divided by the value at T1, then expressed as a percentage (T2/T1%; T3/T1%). The differences in the percentage of recovery between the two groups were compared using a two-sample *t*-test.

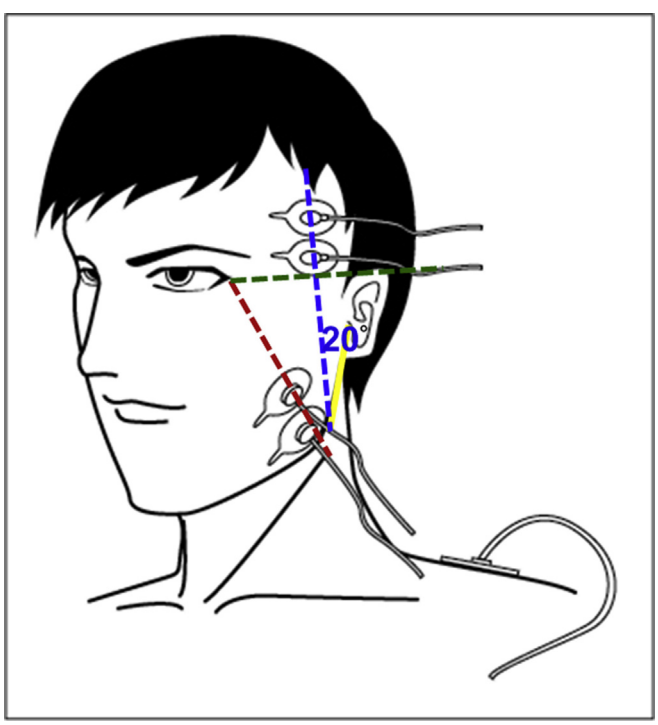

Fig. 1. Positions of the surface electrodes were determined according to the guidelines of Castroflorio et al., 2005. For the masseter muscle, the electrodes are taped along the line from the gonion to the cantus. For the anterior temporalis, two reference lines are required to identify the 20° forward inclination of the posterior mandibular border and the tangent line to the auricle passing through the cantus. The last electrode is placed directly above the intersection of the two lines. The reference electrode is placed at the back of the junction between the neck and shoulder on the left side. Source: Ko et al., 2013.

Table 2
Definitions of measurement variables of myoelectrical activity of masseter and anterior temporalis muscles.

Parameter	Definition
ATR_R	Mean Amplitude of EMG activity of anterior Temporalis muscle in **relaxation** _Right side.
ATR_L	Mean Amplitude of EMG activity of anterior Temporalis muscle in **relaxation** _Left side.
AMR_R	Mean Amplitude of EMG activity of Masseter muscle in **relaxation** _Right side.
AMR_L	Mean Amplitude of EMG activity of Masseter muscle in **relaxation** _Left side.
ATB_R	Mean Amplitude of EMG activity of anterior Temporalis muscle in Biting test of **maximum clenching** _Right side.
ATB_L	Mean Amplitude of EMG activity of anterior Temporalis muscle in Biting test of **maximum clenching** _Left side.
AMB_R	Mean Amplitude of EMG activity of Masseter muscle in Biting test of **maximum clenching** _Right side.
AMB_L	Mean Amplitude of EMG activity of Masseter muscle in Biting test of **maximum clenching** _Left side.
ATb_R	Mean Amplitude of EMG activity of anterior Temporalis muscle in biting test **on the cotton roll** _Right side.
ATb_L	Mean Amplitude of EMG activity of anterior Temporalis muscle in biting test **on the cotton roll** _Left side.
AMb_R	Mean Amplitude of EMG activity of Masseter muscle in biting test **on the cotton roll** _Right side.
AMb_L	Mean Amplitude of EMG activity of Masseter muscle in biting test **on the cotton roll** _Left side.
MATb_R	Maximum Amplitude of EMG activity of anterior Temporalis muscle in biting test **on the cotton roll** _Right side.
MATb_L	**Maximum** Amplitude of EMG activity of anterior Temporalis muscle in biting test **on the cotton roll** _Left side.
MAMb_R	**Maximum** Amplitude of EMG activity of Masseter muscle in biting test_Right side.
MAMb_L	**Maximum** Amplitude of EMG activity of Masseter muscle in biting test **on the cotton roll** _Left side.
MATF_R	Maximum Amplitude of EMG activity of Temporalis muscle in **firing test** _Right side.
MATF_L	Maximum Amplitude of EMG activity of Temporalis muscle in **firing test** _Left side.
MAMF_R	Maximum Amplitude of EMG activity of Masseter muscle in **firing test** _Right side.
MAMF_L	Maximum Amplitude of EMG activity of Masseter muscle in **firing test** _Left side.

A *P*-value <0.05 was considered statistically significant. All the measurement data were analyzed using Statistical Package for Social Sciences software (version 17.0; SPSS Inc., Chicago, IL, USA).

This study was approved by the Institutional Review Board and Medical Ethics Committee at Chang Gung Memorial Hospital (100–1613B). Guidelines of the Helsinki Declaration were followed. Each subject provided informed consent to the investigation after being briefed on its aims and procedures.

3. Results

The mean age and sex distribution of the patients were not significantly different between the control and experimental groups. For the resting tonus in the relaxation test of the anterior temporalis and masseter muscles, both experimental and control group values remained steady from T1 to T3. No statistical difference was found among all values (Table 3).

In the EMG examination of the masseter and anterior temporalis muscles, the amplitude of myoelectrical activity reached 72.6–121.3% of pre-surgical values in the experimental group at T2; in the control group, the amplitude reached 37.5–64.6% at T2 (Table 4).

For the recovery ratio (T2/T1%) of EMG activity of the masseter muscles, all parameters showed a significant difference between the experimental and control groups. The results of all parameters

Table 3
Resting tonus of relaxation (μV) in anterior temporalis and masseter muscle in the experimental and control groups.

Variables	Experimental group						Control group					
	T1		T2		T3		T1		T2		T3	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
ATR_R	7.1	3.9	6.9	3.0	6.9	3.3	7.1	5.3	7.1	5.6	6.3	2.6
ATR_L	7.5	3.2	6.5	2.9	6.5	2.5	7.5	3.3	6.6	4.1	6.1	2.6
AMR_R	3.4	1.2	2.9	1.0	3.7	1.9	4.3	2.8	3.9	2.9	3.3	1.5
AMR_L	3.5	1.1	3.1	0.9	4.1	1.9	4.3	3.1	5.2	4.6	3.4	1.6

No statistical difference was found between groups at same time point, and time point difference (Student's *t*-test) as well as among different time points within same groups (ANOVA). See Table 2 for definition of variables. Unit of measurement: μV.
T1: before surgery; T2: 6 weeks after surgery; T3: 6 months after surgery.

indicated that the early recovery (T2/T1%) of myoelectrical activity in the masseter muscles of patients in the experimental group was better than that of the control group.

Only two parameters of the anterior temporalis muscle showed significant differences between the two groups, these were the ATB_R&L (R and L mean clenching test) and MATF_R (R maximum firing test). The results showed that the average recovery percentages (T2/T1%) of the experimental group were better than those of the control group. Although other parameters did not show a significant difference between the two groups (ATb_R&L, R and L mean cotton roll biting test; MATb_R&L, R and L maximum cotton roll biting test; MATF_L, L maximum firing test), the average recovery percentages (T2/T1%) of the experimental group were still higher than the control group.

At T3, the amplitude of myoelectrical activity of the masseter and anterior temporalis muscles reached 135.1–233.4% of presurgical values in the experimental group. In the control group, the amplitude of myoelectrical activity reached 89.6–122.5% of presurgical values at T3.

For the recovery ratio (T3/T1%) of EMG activity of the masseter muscles, all parameters showed a significant difference between the experimental and control group. The results of all parameters indicated the total recovery (T3/T1%) of myoelectrical activity of the masseter muscles of patients in the experimental group was better than in the control group.

Most of the parameters of the anterior temporalis muscle showed a significant difference between the experimental and the control groups, including ATB_R&L (R and L mean clenching test), ATb_L (L mean cotton roll biting test), MATb_L (L maximum cotton roll biting test), and MATF_R (R maximum firing test). The results of these parameters showed that the average percentages of recovery (T3/T1%) of the anterior temporalis muscle of the experimental group were better than those of the control group. Although some parameters did not show a significant difference between the two groups (ATb_R, R mean clenching test; MATb_R, R maximum cotton roll biting test; MATF_L, L maximum firing test), the average recovery percentages (T3/T1%) of the experimental group were still higher than the control group.

There were no significant differences in the mean recovery ratio of all variables from T2 to T3 (T3 − T2/T1%) between the two groups.

4. Discussion

Surface electromyography was used in this study to measure and record the extent of muscle activity. The muscle potential measured and detected by the EMG device represents a summation of the electrophysiological activity established by multiple motor units around the electrode. Since the EMG reflects muscular activity, the amplitude of the EMG and bite force would be expected

Table 4
Comparison of recovery percentage between the experimental and control groups.

Variables	T2/T1 (%)					T3/T1 (%)					(T3 − T2)/T1 (%)				
	Experimental group		Control group		*P*-value	Experimental group		Control group		*P*-value	Experimental group		Control group		*P*-value
	Mean	SD	Mean	SD		Mean	SD	Mean	SD		Mean	SD	Mean	SD	
ATB_R	96.6	62.7	59.7	48.1	0.006†	189.9	161.2	95.1	68.3	0.003†	43.8	50.1	35.2	25.4	0.295
ATB_L	87.2	63.4	58.9	33.9	0.017*	202.5	146.4	98.6	66.5	0.001†	57.2	65.3	46.5	58.9	0.315
AMB_R	121.3	159.6	52.1	57.4	0.015*	229.0	157.5	105.8	92.8	0.001†	107.7	151.4	71.1	79.7	0.128
AMB_L	102.0	73.7	50.6	55.5	0.002†	233.4	193.3	109.1	77.3	0.001†	131.4	184.4	70.5	70.8	0.051
ATb_R	85.5	81.4	64.6	52.4	0.117	135.2	109.4	94.8	71.9	0.062	49.7	84.5	41.2	58.4	0.340
ATb_L	72.6	63.2	56.1	76.5	0.180	164.1	155.3	91.0	74.6	0.020*	91.5	117.7	53.4	68.8	0.071
AMb_R	80.6	83.6	40.9	31.4	0.005†	174.8	117.9	107.0	78.4	0.007†	90.2	103.1	66.7	64.6	0.155
AMb_L	86.9	69.6	40.3	40.3	0.001†	210.7	171.3	115.2	103.0	0.007†	123.7	145.9	82.3	85.9	0.116
MATb_R	87.2	99.6	63.5	47.9	0.118	136.7	123.1	92.4	71.5	0.061	49.5	92.4	38.1	57.0	0.301
MATb_L	74.2	78.7	54.3	70.5	0.149	176.3	200.8	89.6	81.5	0.026*	92.9	195.9	54.1	96.0	0.193
MAMb_R	85.3	78.3	40.9	33.5	0.003†	173.6	119.5	105.7	80.9	0.008†	88.3	104.3	65.8	63.6	0.165
MAMb_L	83.3	69.3	39.8	41.6	0.002†	195.9	178.7	122.5	112.1	0.042*	112.7	148.1	88.9	94.5	0.252
MATF_R	87.8	53.9	57.9	45.5	0.011*	162.3	127.9	92.9	56.5	0.005†	74.5	110.2	43.9	53.2	0.092
MATF_L	73.9	41.1	58.9	45.4	0.088	135.1	83.3	110.8	111.4	0.180	61.1	82.6	50.6	75.1	0.317
MAMF_R	124.9	198.6	37.5	27.5	0.011*	212.2	211.8	96.8	65.8	0.004†	87.3	125.4	64.3	59.3	0.188
MAMF_L	91.7	86.4	40.4	52.3	0.003†	231.7	241.7	91.0	68.0	0.002†	91.6	89.6	60.9	65.4	0.172

The data analysis used T1 as a reference to compare different time points (T2, T3) for every subject. The T2/T1% represented the effect of early physical rehabilitation on mandibular function. The T3/T1% represented the total recovery of the range of mandibular motion. The (T3 − T2)/T1% indicated the late recovery of the jaw motion variables. See Table 2 for definition of variables.
T1: before surgery; T2: 6 weeks after surgery; T3: 6 months after surgery. See Table 2 for definitions for each of the variables.
*Statistically significant at $P < 0.01$.
†Statistically significant at $P < 0.05$.

to be closely related. If muscle fatigue does not occur, the higher the EMG level, the greater the force assumed to be exerted by the muscle; the extrapolation might not be totally accurate due to the interference of many surrounding factors, including muscle length, muscle anatomy, electrode position and characteristics of contraction filaments (De Luca, 1997; Disselhorst-Klug et al., 2009). Such interference was considered much less in isometric contraction of muscles, with a higher positive correlation between the muscle force and EMG levels. The value of EMG during clenching and biting can be used to estimate the level of masticatory forces if a record of bite force is unavailable.

For class III patients, the myoelectrical activity of the temporalis and masseter muscles is lower than patients with normal occlusion before treatment (Deguchi et al., 1994). The myoelectrical activity gradually increased after OGS correction, but did not reach the level of subjects with normal occlusion until 6–9 months after OGS (Trawitzki et al., 2006). After 3 years follow-up post OGS, the EMG level of the masseter muscle can approach normal levels, but is still at a lower level for the temporalis muscles (Trawitzki et al., 2010). In our study, all the EMG variables had recovered to pre-operative levels by 6 months after OGS. The different results with the study of Trawitzki et al. might be because: 1) intermaxillary fixation was not performed in our study; 2) the amplitude of vibration was dissimilar in the different EMG devices; 3) the placement of the surface electrodes and composition were not identical. Other than these, early physical rehabilitation might be one of the factors. Since there was a wide variation in measurements between individuals, when observing the absolute value of EMG, we applied the pre-operative value as a base line to observe the recovery percentage in this study.

For the outcome of bite force in patients who received sagittal split ramus osteotomy and rigid internal fixation, the molar bite forces could recover to pre-surgical values within 3–6 months of OGS (Kikuta et al., 1994; Kim and Oh, 1997). If the hypothesis of high positive correlation between the force and EMG level is true, we can estimate that the magnitude of bite force could also reach the pre-surgical level at 6 months after OGS.

The EMG value obtained from the relaxation test indicated the resting tonus of the muscles. If the resting tonus was higher than the maximum voluntary contraction in the same muscle, risk of musculoskeletal disorder was strongly suspected (Jonsson, 1978). The resting tonus thus should be at a lower, steady level in healthy muscular performance. In the current study the EMG in the relaxation tests of the experimental group and the control group showed no significant differences. Although the value showed a mild increase in the experimental group and a mild decrease in the control group from T2 to T3, this was not significant. It is suggested that early physical rehabilitation did not change the resting tonus during the period of early physiotherapy. The resting tonus was considered to relate to the vertical facial pattern. The more class III and hyperdivergent the facial type, the higher the resting anterior temporalis activity, and the lesser the increase of clenching masseter muscle activity (Cha et al., 2007). However, we did not classify the facial pattern in this investigation.

For the control group in our study, the recovery percentage of the mean amplitude of EMG activity of the temporalis and masseter muscles in the maximum clenching test (ATB & AMB) was 95.1–98.6% and 105.8–109.1% at 6 months after OGS. The results were similar to the findings of our previous study, in which the amplitude of EMG activity in the masticatory muscles almost recovered to pre-operative levels at 6 months after OGS (Ko et al., 2013). In the current study, in the experimental group at 6 weeks after OGS, the recovery rate measured by the mean amplitude of EMG activity of the temporalis and masseter muscles in the maximum clenching test had already achieved 87.2–96.6% and 102.0–121.3% of pre-surgical levels; these values were significantly higher than in the control group. The results show early physiotherapy could accelerate the recovery of myoelectrical activity and shorten the recovery duration. Existing research mentioned that active or passive muscle stretch and range-of-motion practice could increase maximal mouth opening and maintain mandibular physiological activities (Tuncer et al., 2013). The physiotherapy suggested in this study also includes the daily practice of a range of jaw movements, which trains the masticatory muscles: maximal mouth opening could contract the lateral pterygoid muscle and stretch the jaw elevator muscles; the maximal lateral excursion exercise could induce medial pterygoid muscle contraction on the working side and lateral pterygoid muscle stretching on the balancing side; the maximal protrusive and retrusive exercises can practice lateral pterygoid muscle and temporalis muscle contraction simultaneously; and the active isometric biting exercise uses temporalis, masseter, and medial pterygoid muscle contraction, and could increase maximal biting force and muscular endurance (Thompson et al., 2001). The results of our study indicated that early physiotherapy provides training of the masseter and temporalis muscles, and accelerates the recovery rate of myoelectrical activity after OGS.

When the physiotherapy stopped at the 7th week after OGS, the ongoing recovery percentages (from T2 to T3) of all variables of EMG examination show no significant difference between the two groups. The findings were in line with expectations because the recovery conditions were the same for the two groups from the 7th week to 6 months after OGS. Although the changes in recovery percentages of the two groups were similar after stopping physiotherapy, the experimental group still had better overall recovery percentages than the control group when followed up 6 months later. It seems that subjects who received short-term physiotherapy recovered more quickly and this was sustained at 6 months after OGS. However, even in patients who did not receive early physiotherapy, the recovery of myoelectrical activity of the temporalis and masseter muscles could reach pre-surgical levels at 6 months after OGS according our findings.

In this study, we did not test for masticatory muscle fatigue using the EMG test. Lindström and Hellsing assessed the localized fatigue in masseter muscle during 10 min of clenching. The authors demonstrated a typical gradual decrease in the spectral curve of EMG activity (Lindström and Hellsing, 1983; Mendonça et al., 2005). Buzzinelli and Berzin evaluated the EMG changes in the anterior temporalis and masseter muscles after muscle fatigue induced by continuous chewing. The results showed an endurance time lasting about 500 s and no significant changes on the average amplitude of muscle activation (Buzinelli and Berzin, 2001). Usually jaw elevator muscles are very resistant to fatigue, which is attributed to their good blood supply (Jow and Clark, 1989; Kim et al., 1999; Nakamura et al., 2005) and the high proportion of type I fibers (Sciote et al., 2003). To observe jaw muscle fatigue a long clenching time is required to obtain serial EMG signal changes, and thus to detect signs of fatigue. In our study, the duration required to perform a maximum voluntary contraction (MVC) was 2 s; to perform the firing test, the patient was asked to bite with gradually increasing force to MVC over 10 s. The muscles were allowed to rest between each test. Sustained maximal contraction of masticatory muscles may not have happened in our cases, thus muscle fatigue could be avoided during the EMG test. Torisu and colleagues indicated that jaw muscle fatigue increases resting EMG activity (Torisu et al., 2007). In our study, the muscle resting tonus remained stable and values did not change significantly with time; the condition of muscle fatigue could be ruled out in the methods of EMG testing used.

5. Conclusion

Early physical rehabilitative therapy is helpful for early recovery of myoelectrical activity of the masseter and anterior temporalis muscles after two-jaw surgery in the first 6 weeks. However, after termination of physical therapy, there is no significant difference in the percentage recovery of masticatory muscle activity between the two groups from 6 weeks to 6 months after OGS.

Conflict of interest

The authors have no financial interest to declare in relation to the content of this article.

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